

Pretrained Quantum-Inspired Deep Neural Network for Natural Language Processing

Jinjing Shi¹, Member, IEEE, Tian Chen, Student Member, IEEE, Wei Lai, Student Member, IEEE, Shichao Zhang², Senior Member, IEEE, and Xuelong Li³, Fellow, IEEE

Abstract—Natural language processing (NLP) may face the inexplicable “black-box” problem of parameters and unreasonable modeling for lack of embedding of some characteristics of natural language, while the quantum-inspired models based on quantum theory may provide a potential solution. However, the essential prior knowledge and pretrained text features are often ignored at the early stage of the development of quantum-inspired models. To attacking the above challenges, a pretrained quantum-inspired deep neural network is proposed in this work, which is constructed based on quantum theory for carrying out strong performance and great interpretability in related NLP fields. Concretely, a quantum-inspired pretrained feature embedding (QPFE) method is first developed to model superposition states for words to embed more textual features. Then, a QPFE-ERNIE model is designed by merging the semantic features learned from the prevalent pretrained model ERNIE, which is verified with two NLP downstream tasks: 1) sentiment classification and 2) word sense disambiguation (WSD). In addition, schematic quantum circuit diagrams are provided, which has potential impetus for the future realization of quantum NLP with quantum device. Finally, the experiment results demonstrate QPFE-ERNIE is significantly better for sentiment classification than gated recurrent unit (GRU), BiLSTM, and TextCNN on five datasets in all metrics and achieves better results than ERNIE in accuracy, F1-score, and precision on two datasets (CR and SST), and it also has advantage for WSD over the classical models, including BERT (improves F1-score by 5.2 on average) and ERNIE (improves F1-score by 4.2 on average) and improves the F1-score by 8.7 on average compared with a previous quantum-inspired model QWSD. QPFE-ERNIE provides a novel pretrained quantum-inspired model for solving NLP problems, and it lays a foundation for exploring more quantum-inspired models in the future.

Index Terms—Natural language processing (NLP), neural networks, quantum-inspired neural network, quantum language model, quantum machine learning, sentiment classification, word sense disambiguation (WSD).

I. INTRODUCTION

NATURAL language processing (NLP) [1] has been a subject of study for several years. It proves advantageous for downstream tasks like text classification [2], word sense disambiguation (WSD) [3], knowledge graph [4], and question answering [5]. In recent years, NLP has witnessed substantial advancements, mainly due to the widespread adoption and success of deep learning methodologies, including convolutional neural network (CNN) [6] and recurrent neural network (RNN) [7]. Besides, some pretrained models like BERT [8], GPT [9], and ERNIE [10], which have demonstrated the capability to be fine-tuned for specific tasks, such as sentiment classification [11] and WSD [12]. In 2020, Sadr et al. [13] proposed a multiview deep network for sentiment analysis, which has superior efficiency and generalization performance. In 2022, Sadr and Nazari Soleimandarabi [14] also introduced an attention-based CNN with transfer learning to enhance the performance of sentiment classification. In 2019, Huang et al. [15] proposed three BERT-based models that leverage gloss knowledge to enhance supervised neural WSD, achieve promising performance in the task. In 2021, Barba et al. [16] introduced a transformer-based neural architecture that reformulated WSD as a span extraction problem, attaining a wonderful performance on the English WSD task. In 2023, Pu et al. [17] proposed a lightweight WSD model, which uses the vocabulary knowledge to extract a BERT-based pretraining model. These research results in remarkable gains in model development efficiency and bringing significant performance improvements in NLP. However, NLP still faces two main challenges. The first is the “black-box” problem of neural network models, that millions or even billions of parameters of neural networks are difficult to be explained. The second is that the characteristics of natural language, such as the uncertainty and ambiguity of words, are not embedded in the model [18], which is inconsistent with human perception and may lead to unreasonable modeling. Fortunately, quantum-inspired models [19], [20], [21] that construct neural network model based on quantum theory can effectively improve the

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Jinjing Shi, Tian Chen, and Wei Lai are with the School of Electronic Information, Central South University, Changsha 410083, China (e-mail: shijinjing@csu.edu.cn; tianchen@csu.edu.cn; 1091697485@qq.com).

Shichao Zhang is with the School of Electronic Information, Central South University, Changsha 410083, China, and also with the Key Laboratory of MIMS, College of Computer Science and Technology, Guangxi Normal University, Guilin 541004, China (e-mail: zhangsc@mailbox.gxnu.edu.cn).

Xuelong Li is with the Institute of Artificial Intelligence, China Telecom Corporation Ltd., Beijing 100033, China (e-mail: li@nwpu.edu.cn).

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model interpretability [20], which potentially provides new ideas for overcoming the challenges faced by NLP.

Quantum-inspired models derive their advantages primarily from two aspects. First, the construction of semantic Hilbert space [22] based on quantum probability theory maps natural language (like sememes, words, and sentences) to complex vector space, which enables quantum-inspired model to embed emotion, ambiguity, and other natural language features consistent with human perception. Second, the modules in quantum-inspired models are endowed with physical meanings, like interference, entanglement, collapse, and other quantum properties [23], [24], which may provide conditions for involving interpretable parameters. Moreover, the experimental performance of quantum-inspired model has been verified in many NLP tasks [19], [25], [26].

Recently, researchers have extensively studied the construction of quantum-inspired models and their applications to various NLP scenarios. In 2018, Zhang et al. [25] introduced an end-to-end quantum-inspired language model to deal with question answering tasks, which represents sentence as density matrix and puts forward a joint representation of questions and answers. Based on this model, Li et al. [27] introduced a complex word embedding neural network, which defines the semantic unit in the Hilbert space and uses a complex vector to represent words. In 2019, Li et al. [19] introduced a novel matching model, known as the complex-valued network (CNM), which demonstrated performance levels comparable to traditional CNN and RNN baselines, and Tamburini [28] proposed a quantum-inspired WSD (QWSD) model based on quantum probability theory representing word and sentence in complex domain in the same year. In 2021, Shi et al. [20] proposed ICWE-QNN and CICWE-QNN, improved the quantum-inspired model with gated recurrent unit (GRU), and introduced the attention mechanism and CNN, achieving better experimental results in experiments of text classification. Authentically, above works have made great contributions to the application of quantum theory and promoted the development of quantum-inspired models in NLP. However, their improvements still ignore an important issue that prior knowledge is indispensable for constructing quantum-inspired models, even it has been proved that the models based on quantum probability theory are theoretically suitable for modeling natural language [18]. ELMo [29], Bert [8], GPT [9], and other language models [10], [30], [31] learn text knowledge through pretraining tasks in a large number of corpora, which greatly contributes to their success. While few works embed pretrained text features into quantum-inspired models currently, which may limit the performance improvement of quantum-inspired models.

With the aim of enhancing the performance of the quantum-inspired model, especially for sentiment classification and WSD task, we propose a quantum-inspired pretrained feature embedding (QPFE) method, which can embed additional text features into the model, including emotional features or ambiguity features. Detailedly, the complex embedding layer used in the quantum-inspired model is usually a combination of an amplitude embedding layer and a phase embedding layer. The amplitude layer represents semantic information,

and the phase layer can be used to embed other implicit features, such as the interaction between phrases [27], location information [32], and so on. Further, we pretrain a vanilla quantum-inspired model for specific corpora to obtain the phase parameters containing task-related text features as the initial phase parameters of a new quantum-inspired model, which can extract prior knowledge from text to accelerate the training process of quantum-inspired model and overcome the issue of slow fitting associated with random initialization in previous works [19], [27], [33]. Moreover, we merge the semantic features learned by the pretrained model ERNIE to design a QPFE-ERNIE model to mine rich internal text characteristics, which helps speed up fitting of quantum-inspired model and improve its performance. The effectiveness of QPFE-ERNIE for sentiment classification and WSD is verified in our experiments, and the outcomes demonstrate that our proposed method QPFE and QPFE-ERNIE model can better explore the potential of quantum-inspired model. In general, the contributions can be outlined as follows.

- 1) A QPFE method, including its principle, model construction method, and quantum circuit, is proposed, which is beneficial for embedding more textual features into quantum-inspired model to accelerate the training process and overcome the slow fitting problem in previous quantum-inspired models [19], [25], [27], [33].
- 2) A QPFE-ERNIE model is designed to allow the quantum-inspired model to learn deeper and more complicated text knowledge while retaining the quantum characteristics, which can speed up fitting and enhance the performance of our proposed model.
- 3) The experiments evaluated on sentiment classification and WSD benchmark datasets demonstrate that our QPFE-ERNIE model exhibits a notably superior performance compared to previous quantum-inspired models.

Here is the layout of this article. In Section II, we introduce the preliminary of quantum theory and related works. We describe QPFE, QPFE-ERNIE, and its application in Section III. Our experimental setup and results are detailedly presented in Section IV. And our work is concluded in Section V.

II. RELATED WORK

Quantum theory is introduced in this section, including basic concepts, characteristics, and representations. Further, we review related works concerning word embedding and pretrained models. Table I summarizes the notations used in Preliminary.

A. Preliminary

In quantum physics [34], unlike classical physics, the quantum is the smallest unit that cannot be divided, which is the first property of quantum. For instance, the energy of an electron in an atom only can be one of several values (known as energy levels). The second property of quantum is transition that a quantum system can change its state. For example, electrons in an atom can jump to different energy

TABLE I
NOTATIONS USED IN PRELIMINARY

Notation	Description
$ \psi\rangle$	Quantum superposition state or ket
$\langle\psi $	Quantum superposition state or bra
$\dagger$	Hermitian conjugate operator
ρ	Density matrix
U	Unitary matrix
I	Identity matrix
M_m	Measure operator

levels when they gain or lose energy. The third property is superposition. A particle can be in a superposition state of upspin and downspin. Apart that, quantum states change over time, which calls quantum evolution. A superposition can be measured by a measure operator, after which the superposition state collapses and loses superposition property.

With the mathematical framework of quantum theory, quantum states can be represented by vectors with Dirac notation, marking quantum states with $|\psi\rangle$. Vectors describing quantum states are called state vectors, which can be divided into ket and bra. Ket is column vector and bra is row vector. For a quantum state, its ket and bra are transpose conjugate of each other. In a quantum system, the quantum state is representable through a linear combination of one basis. Unitary transformations can effectively describe the quantum state evolution of an isolated quantum system. Related details about quantum theory are described in the subsequent part of this section.

1) *Quantum State*: A quantum state [35] defined in a Hilbert Space is noted as $|\psi\rangle$ and its conjugate transpose can be noted as $\langle\psi|$. Different from the classical bit, a quantum bit (qubit) has the distinguishing characteristic of existing in a superposition state of $|0\rangle$ and $|1\rangle$, which can be noted as follows:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (1)$$

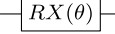
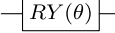
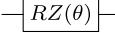
where both α and β are complex numbers and satisfying $|\alpha|^2 + |\beta|^2 = 1$. In addition, $|\alpha|^2$ and $|\beta|^2$ represent the probability of occurrence of states $|0\rangle$ and $|1\rangle$, respectively.

In addition to the state vector representation, the quantum state can also be represented by the density matrix. In quantum mechanism, the density matrix of single particle represented as a pure state ρ can be described as $\rho = |\psi\rangle\langle\psi|$, where ρ satisfies $\rho = \rho^2$. For multiple particles, the total system can be described by a density matrix $\rho = \sum_i w_i |\psi_i\rangle\langle\psi_i|$. Similarly, a word consists of different basic senses, which is viewed as a superposition state. For example, the word “apple” can refer to both the fruit and the Apple company, so the word “apple” is in the superposition of the two meanings when it appears alone. It can be expressed as $|apple\rangle = \alpha|fruit\rangle + \beta|company\rangle$. A sentence is considered as a mixed system for its several words in superposition.

2) *Evolution*: The quantum system always changes its state over time, which is called evolution [35]. The evolution process can be mathematically described as

$$|\psi_2\rangle = U|\psi_1\rangle \quad (2)$$

TABLE II
COMMONLY USED QUANTUM UNITARY OPERATIONS

Unitary operation	Graph representation	Matrix representation
H		$\frac{\sqrt{2}}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
$RX(\theta)$		$\begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$
$RY(\theta)$		$\begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$
$RZ(\theta)$		$\begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$

where U is a unitary matrix satisfying $U^\dagger U = I$, also called quantum gates. The commonly used quantum gates are shown in Table II. H is a Hadamard gate acting on a single qubit. $RX(\theta)$, $RY(\theta)$, and $RZ(\theta)$ are the quantum gates that perform rotations of an arbitrary angle.

A word or phrase changes its meaning based on historical reasons, which is similar with quantum state evolution. For instance, “silly” stems from Old English “saelig” representing happy and careless, while it represents fool nowadays. We can learn that it changes from context information or big corpus, which can be accomplished by RNN, such as GRU [36] and LSTM [37], as well as the pretrained models like BERT and ERNIE.

3) *Quantum Measurement*: Unlike evolution, measurement [38] representing a nonunitary operation is an irreversible process. Quantum measurement is described with a group of measure operators M_m , satisfying $\sum_m M_m^\dagger M_m = I$. Suppose quantum system state is $|\psi\rangle$ before measure operation, and the probability of measured result m can be calculated as follows:

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle. \quad (3)$$

After measuring the system state, the quantum state collapses into

$$|\psi'\rangle = \frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}} \quad (4)$$

and the total sum of the measured probabilities equals 1

$$\sum_m p(m) = \sum_m \langle\psi|M_m^\dagger M_m|\psi\rangle = 1. \quad (5)$$

B. Word Embedding and Quantum-Inspired Word Embedding

Based on the assumption that similar words are situated closely together in the real vector space, Mikolov et al. [39] introduced a novel word embedding method called word2vec. It maps words to vector space integrating semantic information, and the most intuitive case is $vec(\text{“king”}) - vec(\text{“man”}) + vec(\text{“woman”}) = vec(\text{“Queen”})$. Mikolov proposed the training models, CBOW and Skip-gram, depicted in Fig. 1, to acquire word vectors. In the CBOW model, the input consists of the context of a word, and the objective is to make prediction for the missing word. And Skip-gram is the opposite of CBOW, which means giving a word and predicting its context. Pennigton et al. [40] proposed GloVe by combining the learning method proposed in word2vec with statistical information of latent semantic analysis (LSA)

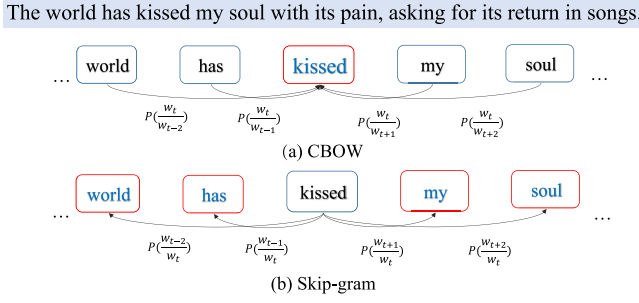


Fig. 1. Two models are proposed to learn word vector representations in word2vec. (a) CBOW model which predicts the current word based on the context. (b) Skip-gram model which predicts word representations surrounding the current word tokens in a sentence.

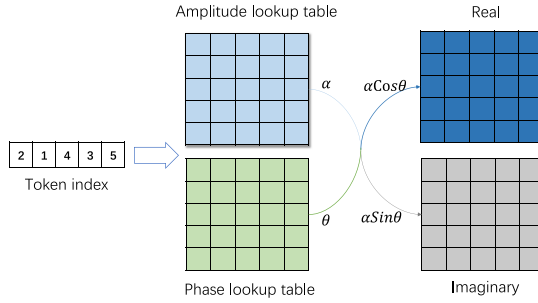


Fig. 2. Process diagram of complex word embedding.

matrix decomposition. GloVe is primarily designed to learn word vectors, whereas word vectors are incidental outcomes of word2vec, whereas word vectors are by-product of word2vec.

Word embeddings represent words as high-dimensional vectors, enabling text to pass through neural networks such as RNN, CNN. There have been many NLP downstream tasks based on word embedding technology making great progress, including text classification [2], machine translation [41], dialog systems [42], etc. Meanwhile, some works have begun to expand word embeddings or use the method for some special directions. For example, Hou et al. [43] introduced a method called joint embedding learning and sparse regression (JELSR) to conduct feature selection. Li et al. [44] introduced a method using auxiliary word embeddings to enhance topic models. Gysel et al. [45] proposed document vector representations for information extraction. Qiu et al. [46] proposed embedding location information to implement recommender systems and Zhang et al. [47] employed embedding techniques to acquire node representations for graph clustering. Xu et al. [48] proposed a dynamic word embedding method that takes the context information into account. Most of the research on word embeddings is still based on the classical perspective. Quantum theory has garnered increasing interest in recent years, particularly in its application to NLP.

Inspired by quantum-based models applied to information extraction tasks, Li et al. [27] developed two quantum models called CE-Sup and CE-Mix to solve sentence classification tasks. CE-Sup treats the sentence as a pure state, while sentence in CE-Mix expressed as mixed state is a linear combination of word quantum states. The complex word embedding method shown in Fig. 2 as the cornerstone of both proposed models mentioned above can capture the implicit meaning of phrases. Compared with the previous Unigram-TFIDF and

fastText bag-of-words models, CE-Sup and CE-Mix exhibit superior performance in the text classification task.

However, the complex word embedding method does not fully harness the distinctive features and potential benefits of quantum theory, since the parameters of the phase layer are initialized randomly, which affects the model performance. The proposed QPFE method, which takes emotional and semantic features as phase initialization parameters, can help solve this problem.

C. Pretrained Models and ERNIE

Pretrained Models: Pretrained models, including ELMo, BERT, GPT, and XLNet, have demonstrated their effectiveness in enhancing the performance of deep learning models on different natural language understanding tasks including sentiment classification [11], named entity recognition [49], question answer [5], recommendation system [50], and so on. RoBERTa [31] improved the performance of BERT by implementing several key modifications, including extended training duration with larger batches, training on more extensive datasets, eliminating the next sentence prediction (NSP) task, introducing dynamic changes in masking patterns, and training on longer sequences. XLNet [30] integrates Transformer-XL and two-stream attention mechanism, as well as employs a permutation language modeling objective, blending the strengths of autoregressive and autoencoding techniques. As a result of this approach, XLNet surpasses BERT in performance across 20 different tasks. BART [51] is trained through the application of an arbitrary noising function on the text, followed by learning a new model to reconstruct the text. This approach has proven to be particularly powerful for text generation tasks and matches the performance of RoBERTa.

ERNIE: ERNIE, or ERNIE 2.0 [10], adopts continual multi-task learning to construct pretraining tasks and train pretrained models, which can extract the lexical, semantic information, and syntactic from training corpora. By utilizing the learned parameters for model initialization, ERNIE simultaneously trains new tasks alongside the original ones. The pretraining tasks encompass word-aware, semantic-aware, and structure-aware pretraining tasks.

For quantum-inspired models, pretrained text features are often ignored, which limits the performance of quantum-inspired models. We build QPFE-ERNIE based on the pretrained model ERNIE, integrating semantic information into quantum-inspired model, under the premise of retaining the characteristics of the quantum-inspired model. Furthermore, we evaluate our model on WSD task, and the experiment results reveal that QPFE-ERNIE outperforms ERNIE. The new features and novelties of our proposed QPFE-ERNIE compared with the existing methods are shown in Table III.

III. PRETRAINED FEATURE EMBEDDING-BASED QUANTUM-INSPIRED DEEP NEURAL NETWORK

Quantum theory has great advantages in expressing uncertainty because of its superposition. This is an essential aspect of natural language modeling, because words have a lot of meanings when not put into context. For an example, the word

TABLE III
NEW FEATURES AND NOVELTIES OF OUR PROPOSED QPFE-ERNIE

	Model	Phase parameters initialization	Pretrained
Xu <i>et al.</i> [48]	Classical	/	Yes
Li <i>et al.</i> [27]	Quantum-inspired	Initialized randomly	No
Ours	Quantum-inspired	Emotional and semantic features	Yes

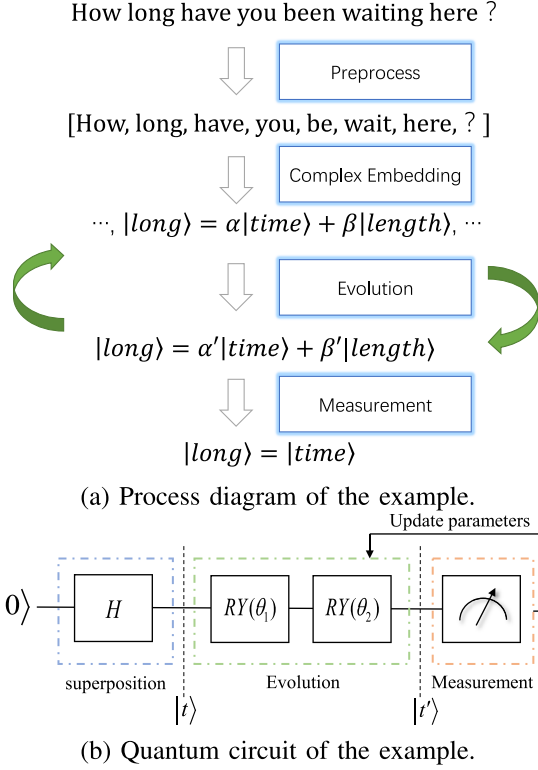


Fig. 3. Example representing word “long” with quantum theory. (a) Process diagram of the example. (b) Quantum circuit of the example.

“long” contains at least two meanings about time and length as shown in Fig. 3(a). The word “long” is represented by one qubit as shown in Fig. 3(b). After superposition, we can obtain $|t\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|0\rangle$ and $|1\rangle$ represent the two meanings “time” and “length” of the word “long,” respectively. Then, the quantum state $|t\rangle$ evolves into the quantum state $|t'\rangle = \alpha'|0\rangle + \beta'|1\rangle$. The parameters of the unitary operations in the evolution process are continuously updated by measurement [52]. Finally, the amplitude of $|0\rangle$ gradually tends to 1, and the amplitude of $|1\rangle$ gradually tends to 0, which means that the meaning of the word “long” collapses to “time.” If it is represented as a quantum superposition, the polysemy can be modeled into network, which is close to the nature of natural language. Next, using contextual information around the word, we can update the probabilities of different meanings. Finally, the whole sentence is regarded as a set of measurement basis for word measurement, and the final meaning of the word is determined.

In the subsequent sections, we elaborate on the specifics of our method. In what follows, we propose QPFE, based on which we merge the semantic features learned by the

pretrained model ERNIE to design a QPFE-ERNIE model and present its application on sentiment classification and WSD.

A. Quantum-Inspired Pretrained Feature Embedding Method

1) *Quantum Representation of Words:* Words in linguistics corresponding to individual particles in a quantum system, as particles existing in upspin and downspin state can be expressed as $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$, have different meaning and emotional significance of words. For example, the word “long” can be described as a distant time noted as $|e_1\rangle$, and it also can represent the relatively large spatial properties noted as $|e_2\rangle$. Therefore, the word “long” in quantum state is expressed as $|long\rangle = \alpha|e_1\rangle + \beta|e_2\rangle$. The norm of α, β represent the probability of the corresponding state. The quantum state of word t can be further formulated as

$$|t\rangle = \sum_i^n w_i |e_i\rangle = \sum_i^n r_i e^{i\theta_i} |e_i\rangle \quad (6)$$

where $|e_i\rangle$ is the i th sense of word t , w_i denotes the weight of $|e_i\rangle$, and r_i and θ_i are amplitude and phase, respectively, as well as n means the number of basic senses. Amplitude $\{r_i\}$ represent semantic information, while phase $\{\theta_i\}$ represent implicit characteristic information, such as emotional information or polysemy information. Ground state $\{|e_i\rangle\}$ are the basic of semantics. In the end, the word is expressed as the superposition state representing uncertainty.

In experiments, the superposition states of words are represented by high-dimensional complex vectors. The complex word vectors are captured by constructing amplitude embedding layer and phase embedding layer. The real and imaginary coefficients of complex words are derived by applying Euler’s formula shown in

$$re^{i\theta} = r(\cos(\theta) + i\sin(\theta)). \quad (7)$$

Word order is with great significance in text processing. Inspired by [32], we define a function that embeds the word order information into the phase, which is shown in

$$f(\theta_{\text{order}}) = e^{i\theta_{\text{order}}} \quad (8)$$

where θ_{order} is the word order information. Thus, words with order information can be represented as

$$\begin{aligned} f(\theta_{\text{order}})re^{i\theta} &= e^{i\theta_{\text{order}}}re^{i\theta} \\ &= r(\cos(\theta + \theta_{\text{order}}) + i\sin(\theta + \theta_{\text{order}})). \end{aligned} \quad (9)$$

From the above equation, we can see that the word order information has been embedded into the phase. Similarly, other information, such as semantic and the interaction between phrases, can be also embedded into the phase.

2) *Pretrained Feature Embedding*: Initial parameters are important for neural network models, especially in the embedding layer. In previous works, the amplitude embedding layer generally uses GloVe as pretrained parameters to represent semantic information, and the phase embedding layer is randomly initialized to $[0, 2\pi]$, whereas the phase layer can represent more complex textual characteristic information such as emotion and polysemy.

Inspired by the pretrained model, we train a vanilla quantum-inspired model on the pretraining task to obtain appropriate initialization parameters. We improve the phase initialization process and assign different initialization parameters to the phase embedding layer for different downstream tasks. In the sentiment classification task, the initialization parameters of phase embedding layer are the features containing emotional information. In the WSD task, the initialization parameters of phase embedding layer are the features containing polysemy information. The features for task embedding can significantly enhance the performance of the model, as substantiated by the experimental results in Section IV.

3) *Features Extraction*: According to quantum theory, the quantum state in an isolated quantum system is changing with time, mathematically expressed as $|\psi'\rangle = U|\psi\rangle$, where U means a unitary matrix. Similarly, certain words have changed their original meanings for sample bias and sociohistorical reasons. Such semantic changes can be learned by training large-scale corpus or inferred from context information. RNN model, such as LSTM and GRU, can learn long memory from input data, which is suitable for processing the sequential data like text. The evolution of quantum states can be simulated using RNN models, which means the quantum states of words vary with the changes of surrounding words.

In fact, the evolution process is to learn sentence-level information by continuously updating the parameters. Specifically, the input is the previous and the current context state, and the GRU outputs a vector containing the information of the previous word through the selective memory gate. The main calculation process of GRU can be simply described as

$$h_t = (1 - z) \odot h_{t-1} + z \odot \tilde{h} \quad (10)$$

where z is an update gate, $\tilde{h}$ denotes the current hidden state, and h_t represents the t -th output. In the above formula, $(1 - z) \odot h_{t-1}$ can forget previous knowledge selectively and $z \odot \tilde{h}$ can learn new knowledge. Therefore, h_t contains information from beginning to the present.

4) *Probabilistic Feature Measurement*: Until now, we have got the quantum states of each word for a sentence, noted as $|t_i\rangle$, and we need to obtain the quantum representation of a whole sentence. In quantum system, multiple particles forming a mixed system usually are expressed as a density matrix $\rho = \sum w_i |\psi_i\rangle\langle\psi_i|$. Also, we represent a sentence as a density matrix $\rho = \sum w_i |t_i\rangle\langle t_i|$, where w_i represents the weight coefficient of word $|t_i\rangle$, in default $w_i = (1/I)$ (I is the length of sentence S). This weight coefficient can also be learned through the attention mechanism for the specific task. After attaining the density matrix representing the sentence, we will perform the sentence measurement operation. First,

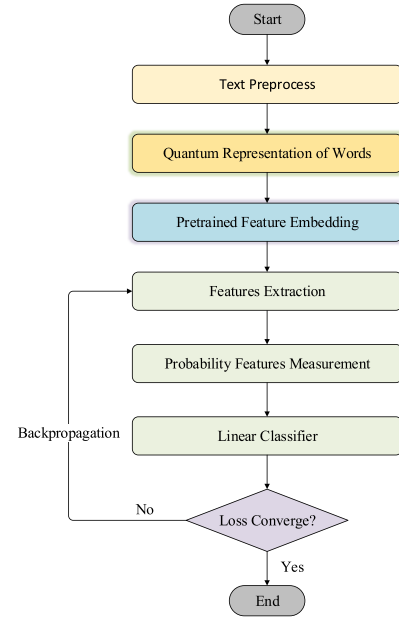


Fig. 4. Total process based on QPFE.

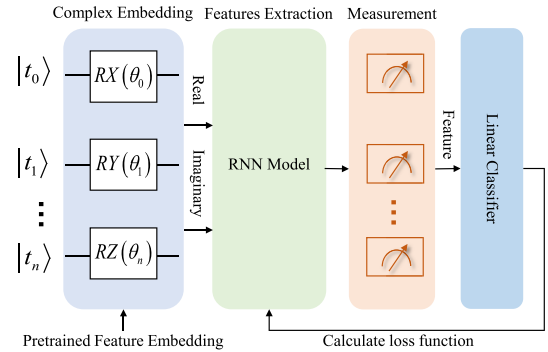


Fig. 5. Schematic quantum circuit diagram of the total process based on QPFE.

we choose a set of measurement basis of $M_m = |\lambda_m\rangle\langle\lambda_m|$ meeting the completeness conditions ($\sum M_m^\dagger M_m = I$). After that, we measure density matrix of sentence with the set of measurement basis. The corresponding measured result is described as follows:

$$p_i = \langle t_i | M | t_i \rangle = \langle t_i | \lambda_m \rangle \langle \lambda_m | t_i \rangle = \langle t_i | \lambda_m \rangle^2. \quad (11)$$

So far, we have completed the construction of the QPFE method and the total process based on QPFE is shown in Fig. 4. In order to understand more intuitively, Fig. 5 gives a schematic quantum circuit diagram of the total task classification process based on QPFE. This process properly integrates quantum operations into classical neural network optimization algorithms, involving the conversion of quantum states to classical data and the iterative update of classical parameters. Compared with previous quantum-inspired models, our method learns prior knowledge from task-related corpora, which is helpful in training loss convergence and improving model performance. In following parts, to enable the quantum-inspired model to acquire deeper and more complicated text knowledge to further optimize the model performance, we

present a QPFE-ERNIE model in the following section, which can achieve better performance in sentiment classification and WSD.

B. QPFE-ERNIE

Classical pretrained models usually consist of stacked Transformer structures, whose parameters are obtained through the pretraining task. We hope to take advantage of these existing NLP techniques to help QPFE learn richer text features. In detail, we integrate the features of QPFE model and ERNIE model to obtain a stronger language model QPFE-ERNIE, which is better than either model at least in sentiment classification and WSD task.

ERNIE [10] is a pretrained model based on Transformer Encoder structure proposed by Baidu, which outperforms BERT and XLNet in 16 NLP downstream tasks. We tried to combine ERNIE with the quantum-inspired model to bolster the model's semantic representation capabilities. ERNIE is used as an independent module to train the high-level features of sentences. Meanwhile, the quantum-inspired model is also trained to obtain the measurement probability features. The two parts of features are concatenated according to the last dimension and the output result is obtained by using the linear layer (corresponding to Fig. 6).

Algorithm 1 gives the algorithm description of QPFE-ERNIE. The general process of constructing the QPFE-ERNIE model mainly includes four steps: 1) quantum state preparation; 2) feature extraction; 3) probability feature measurement; and 4) feature fusion. The details are as follows.

- 1) *Quantum State Representation of Words*: Representing words as a quantum state requires constructing a complex word embedding layer. This layer serves to map discrete words to the Hilbert space conforming to quantum theory. Through this process, pretrained text features are embedded into the embedding layer.
- 2) *Feature Extraction*: Sentence-level features are derived from individual words using GRU. These GRU units facilitate the learning of contextual information within sentences, which is then utilized to update the quantum state representation of words.
- 3) *Probability Feature Measurement*: To perform probability feature measurement, a set of measurement bases are utilized to measure the updated quantum state of words. Subsequently, the measured probability features can be obtained.
- 4) *Feature Fusion*: The parameters of the ERNIE model are first fixed. Following this, tokens are inputted into ERNIE to extract the [CLS] features (a semantic feature vector representing the entirety of the text) as additional semantic information. Then, the [CLS] features are concatenated with the previously derived measured probability features.

RAAdam [53], short for "Rectified Adam," can dynamically adjust the adaptive learning rate based on the variance dispersion of the gradients. One of its key advantages is that it eliminates the need for a manually adjustable parameter for learning rate warm-up. By making full use of the

Algorithm 1 Training Algorithm of QPFE-ERNIE

Input: Processed text tokens: X ; The labels of X : y ;
Output: Predict senses $\hat{y}$;
1: Initialize model parameters, embed pretrained features;
2: Quantum representation of words: $|S\rangle = \mathbf{CE}(X)$;
3: **while** not the Loss L converges **do**
4: Feature Extraction: $|S'\rangle = \mathbf{GRU}(|S\rangle)$;
5: Probability Features: $F_1 = \langle \lambda_m | S' \rangle^2$;
6: Capture extra semantic knowledge: $F_2 = \mathbf{ERNIE}(X)$;
7: Output: $y = \mathbf{Argmax}(\mathbf{Softmax}(W(F_1 \oplus F_2) + b))$;
8: Calculate loss: $L = \mathbf{loss}(\hat{y}, y)$;
9: Backpropagation: optimize parameters with RAdam;
10: **end while**
11: **return** Predict senses $\hat{y}$;

pretrained parameters of ERNIE, the learned features may be complex, and better experimental performance is expected to be achieved.

C. Two Tasks Based on QPFE-ERNIE

We apply QPFE-ERNIE to realize sentiment classification and WSD. For different tasks, we give different inputs to the model and adjust the final classifier structure according to different tasks. Meanwhile, the initialization processes of model parameters for QPFE are different.

1) *Sentiment Classification Based on QPFE-ERNIE*: Sentiment classification, aiming at predicting the sentimental polarity of a sentence, is an elementary text classification task in NLP. We format the context with standard preprocessing steps, including lowercasing letters, splitting words, removing stop words, and encoding tokens into vocabulary indexes. The training labels are converted to "float16" type of paddle tensor. We embed sentiment information into phase embedding layer, where we obtain feature parameters from a Vanilla quantum-inspired model training on sentiment classification datasets.

Fig. 6 shows the detail of QPFE-ERNIE for sentiment classification. The upper and lower parts of the graph are QPFE and ERNIE, respectively, and the features they learn are finally merged together through tensor concatenate operation. The upper part of the figure contains QPFE. First, words representing quantum particle pass through a complex embedding layer composed with an amplitude embedding layer and a phase embedding layer, where the amplitude embedding layer embeds semantic knowledge and the phase embedding layer embeds text features. Next, the real part and imaginary part information are calculated by Euler's formula, and sentence-level features are extracted by GRU. The probability features are obtained through measure operation. The lower part of the figure contains ERNIE, which is composed of ERNIE Embeddings, ERNIE Encoder, and Pooler. ERNIE Embeddings learn the semantics, location, and type information of tokens as well as regularize them. The ERNIE Encoder is a stack of 12 Transformers for learning deep semantic information. Finally, the high-dimensional features are extracted through the pooling layer.

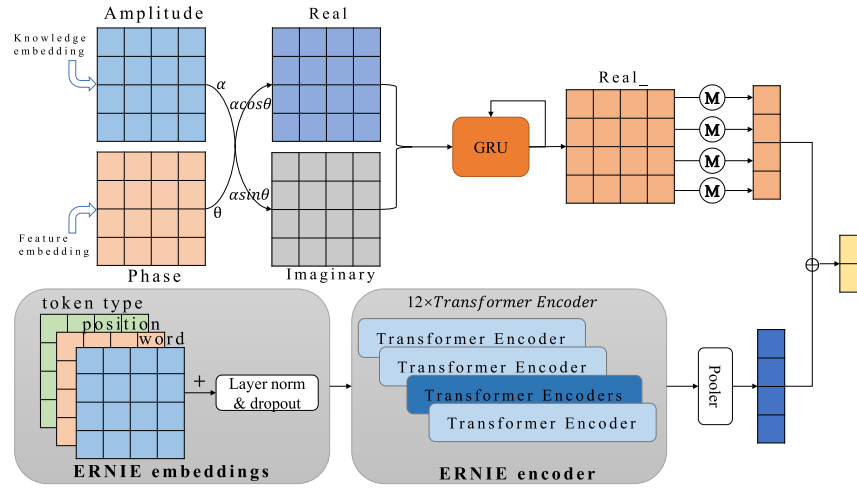


Fig. 6. Architecture of the QPFE-ERNIE model for sentiment classification. $\oplus$ means concatenate two vectors.

Based on the features obtained in previous procedure, the fully connected layer is utilized to map results to vector space and represent the results as predicted probabilities using softmax activation functions. The final prediction results are obtained from the full connection layer

$$\text{Prediction} = \text{Softmax}(WF + b). \quad (12)$$

Here, W corresponds to the weights, while b represents the bias used in the fully connected layer. And F is the fusion features of QPFE and ERNIE.

2) *WSD Based on QPFE-ERNIE*: Compared with sentiment classification, WSD is a more complicated task in data preprocessing and prediction, aiming at finding precise definition of an ambiguous word within a given context from a set of candidate senses. Similarly, we build our QPFE-ERNIE model for WSD. The difference is that we do not regard it as a multiclassification task, but as a question answering task, referring the format of ESC [16]. The input text submitted to the model not only contains the sentence to be predicted but also contains all the glosses of ambiguous words.

Input data: A sentence containing ambiguous word is changed to

$$S = \langle s \rangle w_1, \dots, \langle t \rangle \hat{w} \langle /t \rangle \dots w_n \langle /s \rangle \\ w_1^{d_1}, \dots, w_{|d_1|}^{d_1}, \dots, w_1^{d_k}, \dots, w_{|d_k|}^{d_k} \quad (13)$$

where $\langle s \rangle$ ($\langle /s \rangle$) is begin (end) token of sentence S , and $\langle t \rangle$ ($\langle /t \rangle$) is begin (end) token of ambiguous word $\hat{w}$. $w_i^{d_j}$ is the i th word of gloss d_j . k denotes the number of total glosses.

The model is required to make accurate predictions for the starting position and ending position of glosses associated with ambiguous words. Fig. 7 depicts the QPFE-ERNIE model we built to process WSD. After the formatted text is input into the model, both the QPFE and ERNIE modules simultaneously learn distinct types of features. Our approach embeds both semantic and textual features for the complex word embedding layer of QPFE (obtained by training quantum-inspired model in WSD training datasets). Then, the outputs of the two modules are combined through concatenation, forming the input of the linear classifier. After the classification result passes

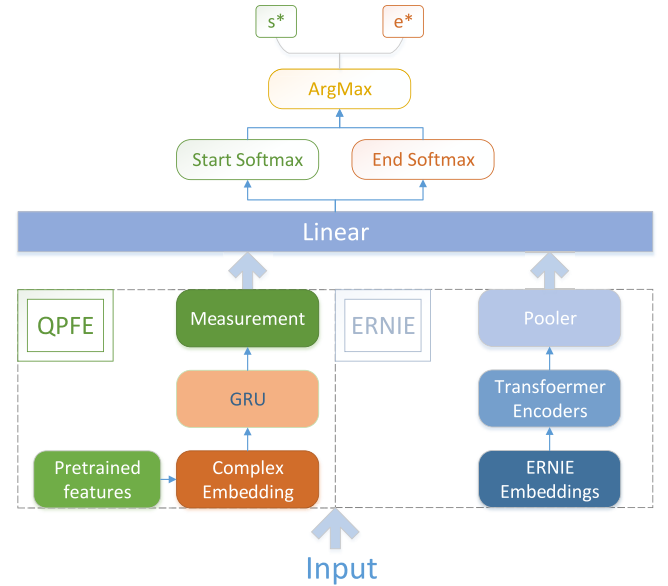


Fig. 7. Architecture of the QPFE-ERNIE model for WSD.

through the two softmax layers and argmax layer, the predicted starting and ending positions are obtained, respectively.

IV. EXPERIMENTS AND EVALUATION

A. Experiments on Sentiment Classification

1) *QPFE-ERNIE for Sentiment Classification*: We build QPFE-ERNIE model for sentiment classification task with paddle framework and run with Tesla V100 in Baidu cloud platform AI Studio.¹ The learning rate is $1e-5$, the number of training epochs is 5, and the batch size is 32.

Datasets: We test our model on five English and one Chinese sentiment datasets. Table IV provides the description of these representative sentiment datasets.

Evaluation Metric: The sentiment classification methods are evaluated using the following metrics.

¹<https://aistudio.baidu.com/>

TABLE IV
DESCRIPTION OF SENTIMENT DATASETS

Dataset	Description	Type	Size
CR [54]	Customer reviews on products	pos/neg	4k
MPQA [55]	Opinion polarity from news	pos/neg	11k
MR [56]	Emotional movie reviews	pos/neg	11k
SST [57]	Stanford sentiment treebank	pos/neg	70k
SUBJ [56]	Subjective reviews	subj/obj	10k
ChnSentiCorp ²	Chinese reviews on hotels	pos/neg	7k

- 1) *Accuracy* is determined by the ratio of correctly predicted instances to all the instances

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (14)$$

- 2) *Precision* is calculated as the ratio of the true positive terms to all positive terms

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (15)$$

- 3) *Recall* is defined as the ratio of true positive terms to all actually true terms

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (16)$$

- 4) *F1-score* is calculated as the harmonic mean of Precision and Recall

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (17)$$

Here, TP represents true positives, TN denotes true negatives, FP represents false positives, and FN denotes false negatives.

Loss function: The cross-entropy loss is utilized in this model, given that sentiment classification is a classical binary classification task. The expression of the loss function is as follows:

$$\text{Loss} = -\frac{1}{n} \sum_i [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]. \quad (18)$$

Let n denote the total number of samples. Each y_i represents the i th label. p_i denotes the probability for positive category prediction.

2) *Comparative Models for Text Classification:* We compare our experiment results to several published works to validate the effectiveness of QPFE-ERNIE for sentiment classification, which are listed as follows.

VanillaGRU consists of an embedding layer, GRU module containing 768 cells and a fully connected layer. We view this model as our baseline since it is a typical model of RNN.

BiLSTM has similar structure as the above model, but the GRU module is replaced by a bidirectional LSTM module, which can learn left and right context information.

TextCNN utilizes CNN along with pretrained word2vec vectors. Specifically, the following experimental results correspond to the CNN-nonstatic model [58].

CE-Mix [59] means Complex Embedding with mixed state. In this model, a sentence is considered as a classical combination of word quantum states.

CNN-Complex-Order [32] extends RNN, CNN, and Transformer, designed to handle complex-valued inputs.

Moreover, it extends word vectors to word functions, enabling the model to capture both the order relationships of words and their global absolute positions.

TextTN [26] introduces a general two-step approach to text classification utilizing Tensor Network. Initially, it employs a generative TN to encode word vectors in the probabilistic space, followed by the classification of text sentences with a discriminative TN.

ERNIE is a prevalent pretrained model designed by Baidu. It is a Transformer-based model and comprises three main components: 1) ERNIE Embeddings; 2) ERNIE Encoder; and 3) Pooler.

3) *Results and Analysis:* The experimental results on five English datasets and one Chinese dataset are presented in Tables V–VIII. VanillaGRU, BiLSTM, TextCNN, and ERNIE are the selected classical baseline models for comparison, while CE-Mix, CNN-Complex-order, and TextTN are the selected quantum-inspired baseline models. In Tables V–VII, we conduct a comparison between our model and the classical baseline models based on their accuracy, F1-score, precision, and recall. In Table VIII, our model is compared with the quantum-inspired baseline models and cited models using the accuracy metric.

Tables V–VII show the metrics (including accuracy, F1-score, precision, and recall) on five English datasets and one Chinese dataset. Our model demonstrates a significant performance improvement compared to GRU, BiLSTM and TextCNN on these datasets in all metrics. For instance, QPFE-ERNIE is higher than TextCNN in accuracy (+15.34), F1-score (+11.33), precision (+18.50), and recall (+2.68). Moreover, QPFE-ERNIE achieves better results than ERNIE in accuracy, F1-score, and precision on two datasets (CR and SST). Compared with ERNIE, QPFE-ERNIE has better accuracy (+1.87), F1-score (+2.15), and recall (+4.49) on MR and has better precision on MPQA (+1.13) and SUBJ (+0.90).

Evaluation on multilingual datasets also demonstrates the benefits of QPFE-ERNIE. In order to verify the cross-language understanding ability of the proposed model, experiments on multilingual dataset are necessary. ChnSentiCorp was used for further evaluation. The ChnSentiCorp dataset is a Chinese-language dataset containing 7k positive and negative reviews of hotels, which is used to verify the sentiment classification performance of the proposed model on a multilingual dataset. QPFE-ERNIE also has better accuracy (+0.83), F1-score (+0.95), and recall (+2.80) than ERNIE on ChnSentiCorp.

From Table VIII, we can conclude that among the quantum-inspired models, our QPFE-ERNIE model exhibits superior accuracy results on three out of the six datasets, i.e., CR (94.7%), MPQA (91.1%), MR (90.0%), and SUBJ (97.6%). QPFE-ERNIE also has the second-best accuracies on SST (94.7%), which are quite near to the best one TextTN (95.3%). Moreover, the accuracy of QPFE-ERNIE is 9.0%, 0.7%, 7.8%, and 2.3% higher than TextTN on CR, MPQA, MR, and SUBJ, respectively. The quantum-inspired models, such as CE-Mix, CNN-Complex-order, and TextTN, are better than the early classical neural network models, have been significantly improved on performance when compared with the classical

TABLE V
EXPERIMENTAL RESULTS OF SENTIMENT CLASSIFICATION ON CR AND MPQA IN FOUR METRICS

Model	CR				MPQA			
	accuracy	F1-score	precision	recall	accuracy	F1-score	precision	recall
VanillaGRU	0.7689	0.8036	0.8128	0.7946	0.8473	0.7538	0.7678	0.7403
BiLSTM	0.7910	0.8301	0.8008	0.8616	0.8417	0.7477	0.7523	0.7433
TextCNN	0.7937	0.8421	0.7704	0.9286	0.8492	0.7531	0.7769	0.7284
ERNIE	0.9418	0.9518	0.9353	0.9668	0.9114	0.8646	0.8357	0.8955
QPFE-ERNIE	0.9471	0.9554	0.9554	0.9554	0.9105	0.8601	0.8488	0.8716

TABLE VI
EXPERIMENTAL RESULTS OF SENTIMENT CLASSIFICATION ON MR AND SST IN FOUR METRICS

Model	MR				SST			
	accuracy	F1-score	precision	recall	accuracy	F1-score	precision	recall
VanillaGRU	0.7233	0.7364	0.7055	0.7701	0.8353	0.8307	0.8528	0.8097
BiLSTM	0.7561	0.7606	0.7495	0.7720	0.8435	0.8387	0.8686	0.8152
TextCNN	0.7664	0.7705	0.7600	0.7813	0.8347	0.8249	0.8753	0.7800
ERNIE	0.8809	0.8801	0.8893	0.8710	0.9434	0.9442	0.9296	0.9523
QPFE-ERNIE	0.8996	0.9016	0.8877	0.9159	0.9473	0.9473	0.9462	0.9483

TABLE VII
EXPERIMENTAL RESULTS OF SENTIMENT CLASSIFICATION ON SUBJ AND CHNSentIcORP IN FOUR METRICS

Model	SUBJ				ChnSentiCorp			
	accuracy	F1-score	precision	recall	accuracy	F1-score	precision	recall
VanillaGRU	0.9090	0.9126	0.9259	0.8996	0.6842	0.6113	0.7801	0.5025
BiLSTM	0.9040	0.9077	0.9219	0.8939	0.8450	0.8473	0.8459	0.8487
TextCNN	0.9210	0.9241	0.9376	0.9110	0.8958	0.8970	0.8992	0.8947
ERNIE	0.9790	0.9802	0.9756	0.9848	0.9425	0.9412	0.9770	0.9079
QPFE-ERNIE	0.9760	0.9771	0.9846	0.9697	0.9508	0.9507	0.9660	0.9359

TABLE VIII
ACCURACY (PERCENTAGE) OF SENTIMENT CLASSIFICATION COMPARED WITH QUANTUM-INSPIRED MODELS

Model	CR	MPQA	MR	SST	SUB
CE-Mix [59]	81.1	86.6	79.8	83.3	92.8
CNN-Complex-order [32]	85.2	90.6	82.5	86.4	95.1
TextTN [26]	85.7	90.4	82.2	95.3	95.3
QPFE-ERNIE (Our model)	94.7	91.1	90.0	94.7	97.6

models except ERNIE. Thus, we specially compared our model QPFE-ERNIE with previous best quantum-inspired models for five datasets, it is derived that QPFE-ERNIE has significantly better experimental results for accuracy on CR (+9.0%), MPQA (+0.5%), MR (+7.5%), and SUBJ (+2.3%) datasets.

B. Experiments on Word Sense Disambiguation

1) *QPFE-ERNIE for Word Sense Disambiguation*: We build QPFE-ERNIE model for WSD using pytorch framework (version 1.7) and train with 3 2080 Ti GPUs. The learning rate is $3e-4$, the number of training epochs is 30, and the batch size is 50.

Datasets: We adopt evaluation suite made by Raganato et al. [60] for English WSD task, including SemCor [61] for training, SemEval-2007 (SE07) [62] for evaluating, and the remaining datasets for testing, i.e.,

TABLE IX
DESCRIPTION OF WSD BASELINE

Dataset		Noun	Verb	Adj	Adv	Total
Train	SemCor	87,002	88,334	31,753	18,947	226,036
Test	SE2	1,066	517	445	254	2,282
	SE3	900	588	350	12	1,850
	SE07	159	296	0	0	455
	SE13	1,644	0	0	0	1,644
	SE15	531	251	160	80	1,022

SemEval-2013 (SE13) [63], SemEval-2015 (SE15) [64], Senseval-2 (SE2) [65], and Senseval-3 (SE3) [66].

Evaluation Metric: The performance evaluation of all the models is conducted using the F1-score mentioned in Section IV-A1 for WSD.

Loss function: We train the model with the target of minimizing the following loss, which indicates interval errors between predicted and correct indices

$$\text{Loss} = \frac{1}{2}(L_s + L_e), \quad (19)$$

$$L_s = \log \sum_{v=1}^l \exp Z_v^s - Z_{i^*}^e, L_e = \log \sum_{v=1}^l \exp Z_v^s - Z_{j^*}^e \quad (20)$$

where s indicates the index position where the prediction begins. e corresponds to the index position of the end of the prediction. v is the traversal variable, from 1 to L (L represents

TABLE X
EXPERIMENTAL RESULTS ON BENCHMARKING WSD DATASETS
TRAINING ON SEMCOR EVALUATED WITH F1-SCORE

Model	SE2	SE3	SE07	SE13	SE15
Classical Models					
BERT	75.9	74.4	68.6	70.6	75.2
ERNIE	74.8	72.6	69.2	75.5	77.5
EWIS [67]	73.8	71.1	67.3	69.4	74.5
GLU [68]	75.5	73.6	68.1	71.1	76.2
SyntagRank [69]	71.6	72.0	59.3	72.2	75.8
SREF _{KB} [70]	72.7	71.5	61.5	76.4	79.5
Generatory [71]	78.0	75.4	71.9	77.0	77.6
GlossBERT [15]	77.7	75.2	72.5	76.1	80.4
Quantum-inspired Models					
QWSD [28]	70.5	69.8	-	69.8	73.4
QPFE-ERNIE	78.8	75.4	72.3	81.0	83.1

¹ - means that the model do not provide this result.

the length of the sentence). i^* and j^* are two special indexes that indicate where the correct answer starts and ends. $Z_{i^*}^e$ and $Z_{j^*}^e$ are the scores associated with the correct start and end indices.

2) *Comparative Models for Word Sense Disambiguation:* We evaluate our method on WSD, and the comparative models are listed as follows.

QWSD [28] is an algorithm for WSD based on quantum probability theory representing word and sentence in complex domain. As far as we know, it is the first quantum-inspired model for WSD task.

Extended WSD incorporating sense embeddings (EWIS) [67] conduct WSD by making predictions over a continuous sense embedding space, rather than using discrete labels, which is able to generalize over both seen and unseen senses.

GLU [68] proposes a pretrained contextualized word representation to capture the context in its hidden vectors, integrating pretrained contextualized word representation into BERT.

SyntagRank [69] utilizes the clarified word pairs in SyntagNet to perform the advanced knowledge-based WSD.

SREF_{KB} [70] proposes a Synset relation-enhanced framework, which uses semantic relations to enhance semantic embedding and implements WSD again with a try-again mechanism.

Generatory [71] leverages a span-based encoding method to fine-tune a pretrained encoder-decoder system.

GlossBERT [15] builds three BERT-based models for WSD and trains models with context-gloss pairs. We show the results of GlossBERT (Sent-CLS-WS) model in the following parts.

3) *Results and Analysis:* Considering WSD task is more difficult and has larger datasets than sentiment classification, we do not compare with simple CNN or RNN models. Instead, we take BERT and ERNIE as comparative models to test the performance of QPFE-ERNIE. Meanwhile, we compare with several famous WSD works, EWIS, GLU, LMMS, GlossBERT, and QWSD. Among them, QWSD is a quantum-inspired model, and the others are classical models. The experiment results are evaluated with F1-score shown in Table X.

The experimental results of QPFE-ERNIE on WSD task are significantly better than the pretrained models BERT (BERT-base) and ERNIE. The F1-score of QPFE-ERNIE is higher than BERT on SE2 (+2.9), SE3 (+1.0), SE07 (+3.7), SE13 (+10.4), and SE15 (+7.9) and the average improvement is +5.2. The F1-score of QPFE-ERNIE is higher than ERNIE on SE2 (+4.0), SE3 (+2.8), SE07 (+3.1), SE13 (+5.5), and SE15 (+5.6) and the average improvement is +4.2. Compared with the previous quantum-inspired model QWSD and other classical models of WSD, it also has better results. Compared with QWSD, the F1-score of QPFE-ERNIE is +8.3 higher on SE2, +5.6 higher on SE3, +11.2 higher on SE13, and +9.7 higher on SE15, resulting in an average F1-score improvement of +8.7.

In summary, our proposed QPFE method relies on complex word embedding representing word in quantum superposition state. In the process of quantum state preparation, we embed task-related features into phase parameters so that the model can learn context internal features easily. In order to enrich semantic information, we build QPFE-ERNIE combining QPFE features with ERNIE features. Finally, QPFE-ERNIE achieves superior performance in WSD task.

V. CONCLUSION

With the objective of tackling the issue of polysemy and improve the performance of quantum-inspired model, QPFE is proposed to embed additional text features into the model, in which evolution process achieved by GRU can learn context information and probabilities features are obtained by measure operation. Based on QPFE, a QPFE-ERNIE model is designed by merging the semantic features learned from the prevalent pretrained model ERNIE, and it is verified with two NLP tasks, sentiment classification and WSD. We perform sentiment classification task with paddle framework and run with Tesla V100 in Baidu cloud platform AI Studio, and we perform WSD using pytorch framework (version 1.7) and train with 3 2080 Ti GPUs. According to the experimental results, the QPFE-ERNIE model is much better than previous quantum-inspired models like CE-Mix, CNN-Complex-order, and TextTN. Especially, it has significant advantages on WSD task comparing with BERT, ERNIE, and previous quantum-inspired model QWSD. It demonstrates a novel neural network modeling method based on quantum theory for carrying out strong performance and great interpretability in related NLP fields. The applications of quantum-inspired models are urgently expected to be investigated in various other domains of NLP, like text summarization, machine translation, dialog systems, retrieval system, recommendation system, etc. In addition, it has the possibility of quantum circuit implementation and pave the way for transformative advances in quantum NLP in the future.

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Jinjing Shi (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Information Science and Engineering, Central South University, Changsha, China, in 2008 and 2013, respectively.

She is currently a Professor with the School of Electronic Information, Central South University. She was supported by the Special Foundation for Distinguished Young Scientists of Changsha and the Natural Science Foundation for Distinguished Young Scholars of Hunan Province in 2019 and 2023, respectively. She has presided over the National Natural Science Foundation Project of China. There are 60 academic papers published in important international academic journals and conferences. Her research interests include quantum computing and quantum machine learning.

Prof. Shi has received the Second Prize of Natural Science and the Outstanding Doctoral Dissertation of Hunan Province in 2015.



Tian Chen (Student Member, IEEE) received the B.S. degree from the School of Computer and Communication Engineering, Changsha University of Science and Technology, Changsha, China, in 2021. She is currently pursuing the master's degree with the School of Electronic Information, Central South University, Changsha.

Her research interests mainly include quantum computing and quantum neural networks.



Wei Lai (Student Member, IEEE) received the B.S. degree from the East China University of Technology, Nanchang, China, in 2020, and the master's degree from the School of Electronic Information, Central South University, Changsha, Hunan, China, in 2023.

His research interests mainly include quantum computing and quantum neural networks.



Shichao Zhang (Senior Member, IEEE) received the Ph.D. degree in computer science from Deakin University, Geelong, VIC, Australia, in 1997.

He is currently a China National-Title Professor with the School of Electronic Information, Central South University, Changsha, China. He is also the Academic Leader of the Key Laboratory of MIMS, College of Computer Science and Technology, Guangxi Normal University, Guilin, China. His research interests include information quality and pattern discovery.

Dr. Zhang served/is serving as an Associate Editor for the *ACM Transactions on Knowledge Discovery from Data*, *IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING*, *Knowledge and Information Systems*, and the *IEEE INTELLIGENT INFORMATICS BULLETIN*. He is a Senior Member of the IEEE Computer Society and a member of ACM.

Xuelong Li (Fellow, IEEE) is the Chief Technology Officer and the Chief Scientist of the Institute of Artificial Intelligence, China Telecom Corporation Ltd., Beijing, China.